

Institute of Actuaries of India

Subject CS1-Actuarial Statistics (Paper A)

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) Correct Answer is **Option C**

It is a basic property of MGFs that for any random variable at $t=0$, the value of the MGF shall be equal to 1. This is not true in case of option C which will return a value of $2/3$, hence it is not valid.

[1]

ii) $M_x(t) = E(e^{tx}) = \int_a^\infty e^{tx} \lambda e^{-\lambda(x-a)} dx$ [1]

$$= \int_a^\infty e^{t\lambda a} \lambda e^{-x(\lambda-t)} dx$$

$$= e^{\lambda a} \lambda e^{-x(\lambda-t)/-(\lambda-t)} \Big|_a^\infty$$

[1]

$$= \lambda/(\lambda-t) e^{at}, \text{ Provided } t < \lambda$$

$$= (1-t/\lambda)^{-1} e^{at}$$

[1]

[3]

iii) $M_x(t) = (1-t/\lambda)^{-1} e^{at}$

$$M'_x(t) = (1-t/\lambda)^{-1} e^{at} a + (1-t/\lambda)^{-2} 1/\lambda e^{at}$$

[1]

$$E(x) = M'_x(0) = a + 1/\lambda$$

[1]

[2]

iv) Correct Answer is **Option B**

The result can be obtained by integrating the pdf from a to x . It returns the following result:

$$- \exp(\lambda a) * [\exp(-\lambda t)]^x_a$$

$$= - \exp(\lambda a) * [\exp(-\lambda x) - \exp(-\lambda a)]$$

$$= \exp(\lambda a) * [\exp(-\lambda a) - \exp(-\lambda x)]$$

[1]

v) $F_x(x) = u = \exp(\lambda a) * [\exp(-\lambda a) - \exp(-\lambda x)]$

$$u / \exp(\lambda a) - \exp(-\lambda a) = - \exp(-\lambda x)$$

$$\exp(-\lambda x) = \exp(-\lambda a) - u / \exp(\lambda a)$$

[1]

Taking logs to the base e on both sides and substituting values of λ and a ,

$$-0.10 * x = \ln(\exp(-0.10 * 0.05) - u / \exp(0.10 * 0.05))$$

$$x = -10 * \ln((1-u) * \exp(-0.10 * 0.05))$$

$$x = -10 * \ln((1-u) * 0.995012)$$

[1]

Using the two values given in the question,

$$@ u = 0.226, x = -10 * \ln((1-0.226) * 0.995012) = 2.61183$$

$$@ u = 0.304, x = -10 * \ln((1-0.304) * 0.995012) = 3.67406$$

[1]

[3]

[10 Marks]

Solution 2:

i) $\Pr(Y < X / 4)$

$$= \Pr(Y < 250)$$

$$= \Pr(3Y < X < \infty, 0 < Y < 250)$$

[0.5]

$$= \int_0^{250} (\int_{3y}^\infty 3/10^6 e^{-x/1000} dx) dy$$

$$= \int_0^{250} (3/10^6 e^{-x/1000} / (-1/1000))_{3y}^\infty dy$$

[1]

$$\begin{aligned}
&= \int_0^{250} 3/10^3 (e^{-3y/1000} - e^{-\infty}) dy \\
&= 3/10^3 (e^{-3y/1000} / (-3/1000))_0^{250} \\
&= (e^0 - e^{-3(250/1000)}) \\
&= 1 - \exp(-0.75) \\
&= 1 - 0.472 \\
&= 0.528
\end{aligned}$$

The probability that the ratio has in fact reduced to 1:4 is 0.528. [0.5]

ii) $f(y) = \int_{x=3y}^{\infty} 3/10^6 e^{-x/1000} dx$ [3]

$$\begin{aligned}
&= 3/10^6 (e^{-x/1000} / (-1/1000))_{3y}^{\infty} \\
&= 3/1000 e^{-3y/1000}
\end{aligned}$$
 [2]

iii) Correct Answer is **Option B**

From part (ii), we can observe that Y is an exponential random variable with the value of parameter $\lambda = 3/1000$. [1]

iv) For random variables X and Y to be independent, $f_X(x) * f_Y(y) = f_{XY}(x, y)$ must be true.

From observation itself, we understand that since $f_{XY}(x, y)$ does not contain any term in y , the above is not true and hence X and Y cannot be independent. [1]

v) Correct Answer is **Option A**

$$\begin{aligned}
&f_X(x) * f_Y(y) \\
&= x/1000^2 e^{-x/1000} * 3/1000 e^{-3y/1000} \\
&= 3x/10^9 * e^{-(x+3y)/1000} \\
&= g_{XY}(x, y) \text{ as mentioned in Option A.}
\end{aligned}$$
 [1]

vi) We have decided to use $g_{XY}(x, y)$ and we know that for this joint probability density function X and Y are independent random variables.

Hence, conditional expectation $E(Y | X > 950)$ is independent of X and hence it is equal to $E(Y)$. [1]

From part (iii), we know that $Y \sim \text{Exp}(3/1000)$.

$$\text{Hence } E(Y | X > 950) = E(Y) = 1000/3 = 333.33$$
 [1]

[2]

[10 Marks]

Solution 3:

i) Correct Answer is **Option A**

All other options represent statements which are not true. [1]

ii) For exponential family, we have write in the form:

$$f(y) = \exp \left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right]$$
 [0.5]

[1]

The given discrete distribution is:

$$\begin{aligned} f(y) &= {}^n C_{ny} * \mu^{ny} * (1 - \mu)^{n-ny} \\ &= \exp (n(y \log \mu + (1 - y) \log(1 - \mu)) + \log ({}^n C_{ny})) \\ &= \exp (n(y \log (\mu / (1 - \mu)) + \log(1 - \mu)) + \log ({}^n C_{ny})) \end{aligned}$$

This is in the form of the exponential family as mentioned above.

Where:

$$\theta = \log (\mu / (1 - \mu)) \quad [0.5]$$

$$\phi = n$$

$$a(\phi) = 1 / \phi \quad [0.5]$$

$$c(y, \phi) = \log ({}^n C_{ny}) \quad [0.5]$$

Using $\theta = \log (\mu / (1 - \mu))$, we can show that $\mu = e^\theta / (1 + e^\theta)$

$$b(\theta) = \log (1 + e^\theta) \quad [1]$$

[4]

$$\text{iii) } E(Y) = b'(\theta) = e^\theta / (1 + e^\theta) = \mu \quad [0.5]$$

$$V(\mu) = e^\theta / (1 + e^\theta)^2 = \mu (1 - \mu) \quad [0.5]$$

$$V(Y) = V(\mu) * a(\phi) = \mu (1 - \mu) / n \quad [0.5]$$

$$E(Z) = n * E(Y) = n * \mu \quad [0.5]$$

$$\text{Var}(Z) = n^2 * V(Y) = n * \mu (1 - \mu) \quad [0.5]$$

This corresponds to the mean $n * \mu$ and variance $n * \mu (1 - \mu)$ of $Z \sim \text{Bin}(n, \mu)$ [0.5]

[3]

iv) Bernoulli variable say X with parameter μ would have the following probability distribution:

$$f(x) = \mu * (1 - \mu) \quad \text{for } x = 0, 1.$$

But the probability density function of Y is

$$f(y, \mu) = {}^n C_{ny} * \mu^{ny} * (1 - \mu)^{n-ny} \quad \text{where } y = 0, 1/n, 2/n, 3/n, \dots, 1$$

Hence Y is not Bernoulli. *Although mean and variance of Y equal to the variance of a Bernoulli distribution, it is not sufficient to conclude that Y is a Bernoulli variable.* [1]

v) Correct Answer is **Option A**

In case of binomial distribution, logit link function is the canonical link function. [1]

[10 Marks]

Solution 4:

i) Correct Answer is **Option D**

All methods are valid methods to estimate the value of parameter based on information from a sample. [1]

$$\text{ii) } \hat{p}_E = \frac{33}{(6 * 7)} = 0.786, \hat{p}_Z = \frac{37}{(11 * 7)} = 0.481, \hat{p} = \frac{70}{(17 * 7)} = 0.588 \quad [1]$$

iii) Correct Answer is **Option A**

let n_i be the total number of questions for the whole batch i , B_i be the total number of correct answer by Batch i .

$$L(b; \theta) = (2\theta)^{BE} (1-2\theta)^{7nE-BE} (\theta)^{BZ} (1-\theta)^{7nZ-BZ} * \text{constant} \quad [2]$$

$$\begin{aligned}
 l(b; \theta) &= \ln L(b; \theta) \\
 &= 33 \ln(2\theta) + (42-33) \ln(1-2\theta) + 37 \ln(\theta) + (77-37) \ln(1-\theta) + \text{constant} \\
 &= 33 \ln(2\theta) + 9 \ln(1-2\theta) + 37 \ln(\theta) + 40 \ln(1-\theta) + \text{constant}
 \end{aligned}$$

iv)

$$\begin{aligned}
 \frac{dl}{d\theta} &= \frac{66}{2\theta} - \frac{18}{1-2\theta} + \frac{37}{\theta} - \frac{40}{1-\theta} \\
 &= \frac{70}{\theta} - \frac{18}{1-2\theta} - \frac{40}{1-\theta}
 \end{aligned}$$

[1.5]

Set equal to zero and solve

$$\begin{aligned}
 \frac{70(1-2\theta)(1-\theta) - 18\theta(1-\theta) - 40\theta(1-2\theta)}{\theta(1-2\theta)(1-\theta)} &= 0 \\
 \Rightarrow 70 - 210\theta - 140\theta^2 - 18\theta + 18\theta^2 - 40\theta + 80\theta^2 &= 0 \\
 \Rightarrow 238\theta^2 - 268\theta + 70 &= 0 \\
 \Rightarrow \theta &= 0.412 \text{ or } 0.714 \\
 \Rightarrow \text{As } \theta < 0.5, \hat{\theta} &= 0.412
 \end{aligned}$$

[1.5]

[1]

[4]

v)

Total number of correct answers	"Elite" Batch	"Zenith" Batch
Method of Moments Estimate	42 * 0.588 = 24.7	77 * 0.588 = 45.3
Method of Maximum Likelihood Estimate	42 * 0.412 = 34.6	77 * 0.412 = 31.7
Observed Values	33	37

[1]

From the table it appears that MLE gives a better fit as predicted values are close to expected values.

[1]

[2]

[10 Marks]

Solution 5:

i) Correct Answer is **Option A**

Since the sample size is small and since the population variance is not known, t distribution would be suitable to perform this test.

[1]

ii)

$$\begin{aligned}
 \bar{X}_A &= 51.48, \quad \bar{X}_B = 40.14, \\
 S_A^2 &= 1/9 * (27804.64 - 10 * 51.48^2) = 144.7484 \\
 S_B^2 &= 1/9 * (18215.88 - 10 * 40.14^2) = 233.7427
 \end{aligned}$$

[1.5]

Pooled Variance:

$$\begin{aligned}
 S_p^2 &= 1/18 * (9 * 144.7484 + 9 * 233.7427) \\
 &= 189.2456
 \end{aligned}$$

[1]

95 % confidence interval is: $(\bar{X}_A - \bar{X}_B) \pm t_{n_A+n_B-2} * S_p^2 * \sqrt{\frac{1}{n_A} + \frac{1}{n_B}}$

$$\begin{aligned}
 &= (51.48 - 40.14) \pm 2.101 \sqrt{189.2456} \sqrt{\frac{2}{10}} \\
 &= (-1.58569, 24.26569)
 \end{aligned}$$

[1]

Since 0 lies in the above confidence interval, we can conclude that at 5% level of significance, there is insufficient evidence to reject the null hypothesis.

[0.5]

[4]

iii) Correct Answer is **Option D**

Normality of the population data and equal population variances are the assumptions used for conducting a t-test. [1]

iv) Sample size:

The width of the confidence interval is

$$2 * t_{2.5\%, 2n-2} \sqrt{\frac{2}{n}} \sqrt{\frac{144.7484(n-1) + 233.7427(n-1)}{2n-2}}$$

$$= t_{2.5\%, 2n-2} \frac{38.9097}{\sqrt{n}}$$

[1]

This should be less than 20, so using percentage points of the t distribution,

We have:

$$n = 15 \Rightarrow t_{2.5\%, 2n-2} = 2.048$$

$$\Rightarrow 38.9097 * 2.048 / \sqrt{15} = 20.57511 > 20$$

[1]

$$\text{And } n = 16 \Rightarrow t_{2.5\%, 2n-2} = 2.042$$

$$\Rightarrow 38.9097 * 2.042 / \sqrt{16} = 19.8634 < 20$$

Hence the minimum sample size required is 16. Hence, at least 6 additional circles must be subject to a test run before going ahead with full blown implementation.

[1]

[3]

v) At this new width, confidence interval would be:

$$(11.34 - 19.8634/2, 11.34 + 19.8634/2)$$

$$= (1.408, 21.272)$$

[0.5]

Since this confidence interval does not contain 0, we can conclude that there is sufficient evidence to reject the null hypothesis.

[0.5]

[1]

[10 Marks]

Solution 6:

i) Correct Answer is **Option B**

Since the data is perfectly monotonically decreasing, both the coefficients would be equal to -1 (perfect negative correlation).

[2]

ii) $H_0: \rho = 0$
 $H_1: \rho > 0$

[0.5]

Under H_0 , the sampling distribution of Kendall's rank correlation coefficient is approximately normal with mean 0 and variance = $2(2n+5) / 9n(n-1)$

[0.5]

$$\text{Variance} = 2(20+5)/90(9) = 50/810 = 0.061728$$

[0.5]

Observed value of the test statistic

$$= (r_k - \text{mean}) / \sqrt{\text{variance}}$$

$$= (0.11 - 0) / \sqrt{0.061728}$$

$$= 0.4427$$

[1]

This does not exceed the upper 0.05% point of the standard normal distribution (3.2905).

[0.5]

So we do not have sufficient evidence to reject the null hypothesis.

[0.5]

[0.5]

Hence, we can conclude that the inflation rates for Freedonia and Genovia are not positively correlated.

[4]

iii) The completed table with the rank values for Genovia is given below:

Rank Freedonia	Rank Genovia	Number of Concordant Pairs	Number of Discordant Pairs
1	6	4	5
2	3	6	2
3	1	7	0
4	4	5	1
5	8	2	3
6	10	0	4
7	9	0	3
8	7	0	2
9	2	1	0
10	5	-	-
		25	20

In case of Rank 1 (Freedonia), there are 4 concordant pairs and 5 discordant pairs. So, the rank value here for Genovia is higher than 4 rank values (10, 9, 8, 7) but lower than 5 rank values (1, 2, 3, 4, 5). It must be 6.

In case of Rank 2 (Freedonia), there are 6 concordant pairs and 2 discordant pairs. So, the rank value here for Genovia is higher than 6 rank values (10, 9, 8, 7, 5, 4) but lower than 2 rank values (1, 2). It must be 3. Kindly note that 6 is already considered in the upper cell and it is not being taken into consideration.

The above process can be continued till we get all rank values for Genovia.

[2.5]

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)}$$

[0.5]

$$\sum d^2 = -5^2 + -1^2 + 2^2 + 0^2 + -3^2 + -4^2 + -2^2 + 1^2 + 7^2 + 5^2 = 134$$

[1]

$$r_s = 1 - 6 \cdot 134 / 10(100 - 1) = 0.1879$$

[1]

[5]

iv) If we want to retain those components which explain 90% of the total variance, PC1 should be retained as it accounts for 95.88% of the total variance.

[1]

Based on the Scree Plot, the plot becomes flat from PC2 and onwards. Hence using Scree Test, PC1 should be retained as variances level off after PC1.

[1]

As per Kaiser's Test only those PCs with variances greater than 1 should be retained (applicable in case of scaled data). Since, only PC1 has variance greater than 1, only PC1 should be retained.

[1]

[3]

v) Correct Answer is **Option C**

Principal components are un-correlated linear combinations of the variables of the original data set. Hence correlation coefficient between PC1 and PC2 will be equal to 0.

[1]

[15 Marks]

Solution 7:

i) Correct Answer is **Option B**

Since we are modelling N which represents the number of trials to be performed until the first success occurs, the appropriate distribution would be geometric distribution. [1]

ii) The prior distribution of " p " is uniform over the interval $[0,1]$

$$\text{So } f_{\text{prior}}(p) = 1 \quad 0 \leq p \leq 1 \quad [0.5]$$

Sample contains only one observation n_1 . So the likelihood function of " p " is:

$$L(p) = P(N = n_1) = (1 - p)^{(n_1-1)} * p$$

The above expansion is based on the fact that $N | p \sim \text{Geometric}(p)$ [1]

Combining the prior PDF and the likelihood function, we get,

$$f_{\text{posterior}}(p) \propto f_{\text{prior}}(p) * L(p)$$

$$f_{\text{posterior}}(p) \propto (1 - p)^{(n_1-1)} * p$$

$$f_{\text{posterior}}(p) \propto (1 - p)^{(n_1-1)} * p^{(2-1)} \quad \text{for } 0 \leq p \leq 1 \quad [1]$$

So the posterior distribution of p is Beta ($2, n_1$) [0.5]

[3]

iii) Prior mean of p under $U[0,1]$ is given by:

$$E(p) = (0+1)/2 = 0.50 \quad [0.5]$$

We have to calculate probability that P exceeds 0.50 using the posterior distribution of P .

$$P(P > 0.50)$$

$$= \int_{0.50}^1 (1 - p)^{(n_1-1)} * p \, dp$$

$$= \int_{0.50}^1 (1 - p)^{(3-1)} * p \, dp$$

$$= \int_{0.50}^1 (1 - p)^2 * p \, dp$$

$$= \int_{0.50}^1 (1 - 2p + p^2) * p \, dp$$

$$= \int_{0.50}^1 (p - 2p^2 + p^3) \, dp$$

$$= (1^2 - 0.50^2) / 2 - 2/3 * (1^3 - 0.50^3) + 1/4 * (1^4 - 0.50^4)$$

$$= 0.026042 \quad [2.5]$$

This signifies that there is only 2.6% chance that the vote share of XJP will cross 50% (which is the prior expectation based on historical vote shares). [1]

[4]

iv) Likelihood function now is given by:

$$L(p)$$

$$= P(N_1 = n_1) * P(N_2 = n_2) * P(N_3 = n_3) * \dots * P(N_{50} = n_{50})$$

$$= (1 - p)^{(n_1-1)} * p * (1 - p)^{(n_2-1)} * p * (1 - p)^{(n_3-1)} * p * \dots * (1 - p)^{(n_{50}-1)} * p$$

$$= (1 - p)^{(n_1+n_2+\dots+n_{50}-50)} * p^{50}$$

$$= (1 - p)^{(200-50)} * p^{50}$$

$$= (1 - p)^{150} * p^{50} \quad [1.5]$$

Hence, posterior distribution of p is given by:

$$f_{\text{posterior}}(p) \propto (1 - p)^{150} * p^{50}$$

$$f_{\text{posterior}}(p) \propto (1 - p)^{(151-1)} * p^{(51-1)} \quad \text{for } 0 \leq p \leq 1 \quad [1]$$

So the posterior distribution of p is Beta ($51, 151$) [0.5]

[3]

- v) Bayesian estimate of “p” under squared error loss is the mean of the posterior distribution which is given by:

$$E(P | n) = \alpha / (\alpha + \beta) = 51 / (51 + 151) = 0.2525 \quad [1]$$

$$E(P | n) = Z * \text{sample mean} + (1 - Z) * \text{prior mean}$$

$$0.2525 = Z * (50/200) + (1 - Z) * 0.50$$

$$0.2525 = 0.25Z - 0.50Z + 0.50$$

$$0.25Z = 0.2475$$

$$Z = 0.99$$

So, if we want to estimate the posterior mean as a credibility estimate,

$$E(P | n) = 0.99 * \text{sample mean} + 0.01 * \text{prior mean} \quad [2]$$

[3]

- vi) Correct Answer is **Option C**

200 is a very small sample size to draw any concrete conclusion. Increasing the sample size would be the ideal way forward.

[1]

[15 Marks]

Solution 8:

- i) Correct Answer is **Option B**

Value of estimates for the intercept, X_1 and X_2 will be the values of α , θ_1 and θ_2 respectively. [1]

- ii) Adjusted $R^2 = 1 - (n - 1) / (n - k - 1) * (1 - R^2)$ [1]

We have $n = 10$ and $k = 2$ predictors and $R^2 = 0.9005$ [0.5]

Adjusted R^2

$$= 1 - (10 - 1) / (10 - 2 - 1) * (1 - 0.9005)$$

$$= 87.21\%$$

[0.5]

[2]

- iii) $\hat{y}$ Emerald City
 $= 47.5601 + 0.6924 (125) - 0.6624 (90)$
 $= 74.4941$

[0.5]

$\hat{e}$ Emerald City

$$= 77 - 74.4941$$

$$= 2.5059$$

[0.5]

$\hat{y}$ Dark City

$$= 47.5601 + 0.6924 (124) - 0.6624 (72)$$

$$= 85.7249$$

[0.5]

$\hat{e}$ Dark City

$$= 88 - 85.7249$$

$$= 2.2751$$

[0.5]

[2]

- iv) $S_{yy} = 776.10$
 $S_{zz} = 1472.10$
 $S_{yz} = 1013.90$

[1]

$\hat{\mu}$

$$= S_{yz} / S_{zz}$$

[1]

$$= 1013.90 / 1472.10$$

$$= 0.6887$$

$$\hat{\lambda}$$

$$= \bar{y} - \hat{\mu} * \bar{z}$$

$$= 80.30 - 0.6887 * 43.70$$

$$= 50.2019$$

[1]

$$\hat{y}_{\text{Emerald City}}$$

$$= 50.2019 + 0.6887 (125-90)$$

$$= 74.3064$$

[0.5]

$$\hat{e}_{\text{Emerald City}}$$

$$= 77 - 74.3064$$

$$= 2.6936$$

[0.5]

$$\hat{y}_{\text{Dark City}}$$

$$= 50.2019 + 0.6887 (124-72)$$

$$= 86.0143$$

[0.5]

$$\hat{e}_{\text{Dark City}}$$

$$= 88 - 86.0143$$

$$= 1.9857$$

[0.5]

[5]

v)

$$R^2$$

$$= S_{xz}^2 / (S_{xx} * S_{zz})$$

$$= (1013.90)^2 / (776.10 * 1472.10)$$

$$= 89.9778\%$$

[1]

$$\text{Adjusted } R^2$$

$$= 1 - (10 - 1) / (10 - 1 - 1) * (1 - 0.899778)$$

$$= 88.725\%$$

[1]

[2]

vi)

Correct Answer is **Option D**

$$2 = (0.6887) * Z_{\text{reduction}}$$

$$Z_{\text{reduction}} = 2 / 0.6887 = 2.90$$

[1]

vii)

a) We are given that $\bar{W} = 0$.

$$S_{ww} = \sum (w - \bar{w})^2 = \sum (w - 0)^2 = \sum (z - \bar{z})^2 = S_{zz}$$

[1]

$$S_{yw} = \sum (y - \bar{y})(w - \bar{w}) = \sum (y - \bar{y})(w - 0) \quad \sum (y - \bar{y})(z - \bar{z}) = S_{yz}$$

[1]

$$\hat{E} = S_{yw} / S_{ww} = S_{yz} / S_{zz} = \hat{\mu}$$

[1]

$$\text{b) } \hat{\delta} = \bar{y} - \hat{\mu} * \bar{w} = \bar{y} - \hat{\mu} * 0 = \bar{y} = \hat{\lambda} + \hat{\mu} * \bar{z}$$

[1]

[4]

viii)

City	Multiple Linear Regression Model	Bivariate Linear Model	Improved Bivariate Linear Model
Emerald City ($\hat{y}, e$)	(74.49, 2.51)	(74.31, 2.69)	(74.31, 2.69)
Dark City ($\hat{y}, e$)	(85.72, 2.28)	86.02, 1.99)	(86.02, 1.98)
Adjusted R^2	87.21%	88.73%	88.72%

[1]

In terms of the predicted responses and residuals, for Emerald City, the multiple linear regression model appears to be a better fit. However for Dark City, the bivariate linear model gives better results as compared to the multiple linear regression model.

However in terms of Adjusted R^2 (which measures the variation of the predicted responses to actual responses), Bivariate Linear Model appears to be a better fit.

[1]

Improvised Bivariate Linear Model just employs a linear combination of the explanatory variable of the Original Bivariate Linear Model and hence gives almost similar results like the original model. Unlike the presumption made by your son, it is clear that the improvised model does not provide a better fit as compared to the original model.

[1]

[3]

[20 Marks]
